



**Universität
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Veech's Examples of Non-Ergodic IETs: from Renormalization Perspective

Master Thesis in Mathematics

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Abstract

We revisit Veech's 1969 examples of minimal but non-uniquely ergodic interval exchange transformations using the framework of renormalization in dynamical systems. In particular, we used the Ostrowski expansion and its dynamical interpretation to help us study the irrational rotations with a marked point. The criterion used for the non-ergodicity related to the existence of solutions to some cohomological equations.

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1 Introduction

Let $I = [0, 1)$. An interval exchange transformation (or IET for short) $T : I \rightarrow I$ is a piecewise isometry of the interval with finitely many discontinuities (see § 1.1 for the formal definition). Also see figure 1 below, where we mean that each subinterval on the upper part of the figure is mapped to the subinterval with the same label on the lower part of the figure, by translation. Notice that IETs always preserve the Lebesgue measure on the interval.

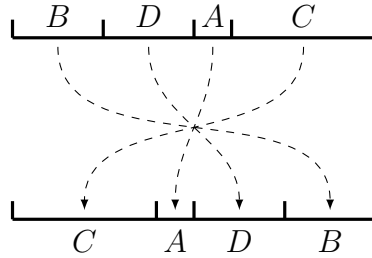


Figure 1: An example of IET

While IETs are fascinating in their own right as simple yet rich dynamical systems, much of their modern importance stems from their deep and natural connection with translation surfaces, that is, closed surfaces with an atlas whose coordinate changes are all translations of the plane, outside a finite number of conical singularities. A translation surface comes with a flat Riemannian metric, transported from the plane through local charts, and the geodesic flow is meant with respect to this metric. The IETs appears as first-return maps of geodesic flows on such surfaces (See [Yoc06] for a precise formulation and see also 1.1.2 for examples of our purpose).

Keane [Kea75] proved that, for all but a countable set of choices of the subintervals, the interval exchange transformation is minimal: every orbit is dense. From this one can recover the minimality of the geodesic flow discussed previously. In the same paper, he also conjectured that all minimal IETs are unique ergodic: the multiples of Lebesgue measure are the only invariant measures. However, various counterexamples were subsequently constructed: Veech [Vee69], which is realized by Keynes-Newton [KN76], Keane himself [Kea77], Katok [Kat73], and Sataev [Sat75]. Veech's example is later been interpreted in terms of flows on translation surface by Masur and Smillie [MS91].

In recent decades, many constructions of examples or counterexamples in the study of Teichmüller dynamics and area-preserving flows on surfaces originating from Veech's example (see, e.g., [CT17, CW19, FKZ25]). That is one of the reasons we thought it worth revisiting this famous example.

The main focus thesis lies on Veech's examples, which is an extension of a rotation of angle α by the group $\mathbb{Z}/2\mathbb{Z}$ by marking a point β and going from one copy of the torus to the other one on the interval $[0, \beta)$. It will be shown in 1.1.1 that these are indeed IETs. What has been shown in Veech's paper [Vee69] is the following:

Theorem 1.0.1 (Veech). *For α with unbounded partial quotients, there are uncountably many β to make the $\mathbb{Z}/2\mathbb{Z}$ extension minimal but not uniquely ergodic.*

The original proof of Veech is generally analytic and arithmetic, based on the study Ostrowski expansion (see 2.3 for formal treatment) of β with respect to α . The present thesis is devoted to a detailed exposition of Veech's examples from the viewpoint of renormalization theory. While the example is well known, a self-contained treatment that systematically highlights the role of renormalization – here materialized through the continued fraction algorithm and the associated Ostrowski expansion – is largely absent from the literature, as far as we know. We shall consider a slightly more generalized Veech's examples, which are $\mathbb{Z}/p\mathbb{Z}$ -extensions of rotations with the same step function taking the two values 0 and 1. This case of extensions of rotations were already studied by Veech [Vee75], Stewart [Ste81], Merrill [Mer85] and more recently Ferenczi and Hubert [FH24].

1.1 Background materials and the main result

Let $I = [a, b) \subseteq \mathbb{R}$ be a bounded semi-open interval, and let $\mathcal{A}$ be a finite alphabet consisting of $d \geq 2$ symbols. A partition of I into subintervals indexed by $\mathcal{A}$ is denoted by $\{I_\alpha : \alpha \in \mathcal{A}\}$. An *interval exchange transformation* (IET for short) is a bijection $T : I \rightarrow I$ that acts as a translation on each subinterval I_α . Such a map, denoted by $T_{\pi, \lambda}$, is uniquely determined by its **combinatorial** and **metric** data as follows:

- (i) **Combinatorial data:** A pair $\pi = (\pi_0, \pi_1)$ of bijections $\pi_\varepsilon : \mathcal{A} \rightarrow \{1, \dots, d\}$ ($\varepsilon \in \{0, 1\}$) that describes the ordering of the subintervals I_α before and after the application of T . Conventionally, this is represented by the matrix

$$\pi = \begin{pmatrix} \alpha_1^0 & \alpha_2^0 & \cdots & \alpha_d^0 \\ \alpha_1^1 & \alpha_2^1 & \cdots & \alpha_d^1 \end{pmatrix},$$

where $\alpha_j^\varepsilon = \pi_\varepsilon^{-1}(j)$ for $\varepsilon \in \{0, 1\}$ and $j \in \{1, 2, \dots, d\}$.

- (ii) **Metric data:** A vector $\lambda = (\lambda_\alpha)_{\alpha \in \mathcal{A}}$ with strictly positive entries, where $\lambda_\alpha = |I_\alpha|$ denotes the Lebesgue length of the subinterval I_α .

By definition, the endpoints of the subintervals I_α typically constitute the set of discontinuities for the IET. We say that $T_{\pi, \lambda}$ satisfies the *Keane condition* if the infinite orbit of every unique discontinuity point remains disjoint from the set of all discontinuities. A classical result [Kea75] states that any IET satisfying the Keane condition is minimal.

1.1.1 Veech's Example

Let p be a prime, $\alpha \in (0, 1) \setminus \mathbb{Q}$ and let $R_\alpha : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$ be the irrational rotation defined by $R_\alpha x = x + \alpha \pmod{1}$. For a given parameter $\beta \in (0, 1)$, we consider the $\mathbb{Z}/p\mathbb{Z}$ -extension of R_α , denoted by $T_{\alpha, \beta}$:

$$T_{\alpha, \beta} : \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \longrightarrow \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}, \quad (x, j) \longmapsto (R_\alpha x, \mathbf{1}_{[0, \beta)}(x) + j).$$

We endow the space $\mathbb{R}/\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ with the product measure $\mu \times m$, where μ is the standard Lebesgue measure on $\mathbb{R}/\mathbb{Z}$ and m is the Haar measure on $\mathbb{Z}/p\mathbb{Z}$.

Lemma 1.1.1. *The skew-product $T_{\alpha, \beta}$ is conjugate to an IET on $3p$ intervals.*

Proof. For brevity, we restrict our proof to the case $1 - \alpha > \beta > \alpha$; the remaining cases follow from a completely analogous argument. Consider the map

$$\Phi : \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \rightarrow [0, p), \quad \Phi(x, j) = x + j.$$

Let $S = \Phi \circ T_{\alpha, \beta} \circ \Phi^{-1}$. We partition the interval $[0, p)$ into $3p$ subintervals as follows:

$$\begin{aligned} I_1^{(j)} &= [j, j + \beta), \\ I_2^{(j)} &= [j + \beta, j + 1 - \alpha), \quad \text{where } j \in \{0, 1, \dots, p-1\}. \\ I_3^{(j)} &= [j + 1 - \alpha, j + 1), \end{aligned}$$

A direct calculation shows that on each subinterval, S acts as a pure translation:

$$S(x) = \begin{cases} x + \alpha + 1, & x \in I_1^{(j)}, 0 \leq j \leq p-2 \\ x + \alpha, & x \in I_2^{(j)}, 0 \leq j \leq p-1 \\ x + \alpha - 1, & x \in I_3^{(j)}, 0 \leq j \leq p-1 \\ x + \alpha - p, & x \in I_1^{(p-1)} \end{cases}$$

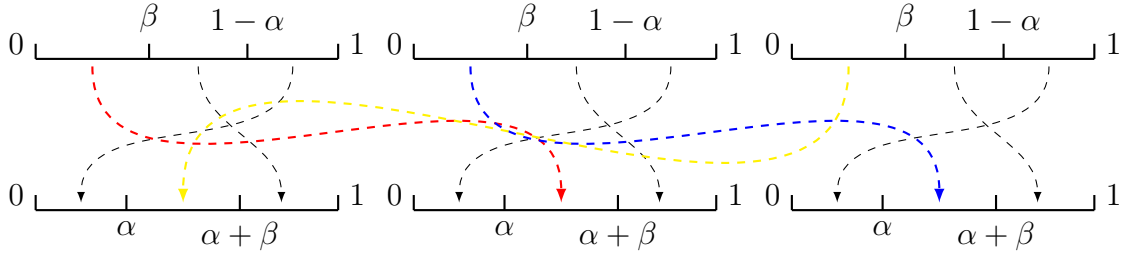


Figure 2: Illustration of the proof of Lemma 1.1.1 in the case $1 - \alpha > \beta > \alpha$, $p = 3$

Hence, S is a $3p$ -IET, completing the proof. ■

Proposition 1.1.2. *If $\beta \notin \mathbb{Z}\alpha \pmod{1}$, then $T_{\alpha, \beta}$ is minimal.*

Proof. If $\beta \notin \mathbb{Z}\alpha \pmod{1}$, the $3p$ -IET construction obtained in the previous lemma satisfies the Keane condition, which immediately implies the minimality of $T_{\alpha, \beta}$. ■

1.1.2 Geometric model

It is also worth mentioning a natural geometric model for Veech's example here, which appeared in many applications of Veech's examples, e.g. [MS91, CT17].

Define the lattice given by α

$$\Lambda_\alpha := \begin{pmatrix} 1 & -\alpha \\ 0 & 1 \end{pmatrix} \mathbb{Z} \oplus i\mathbb{Z}$$

Let

$$(X_{\alpha,\beta}, \omega) = (\mathbb{C}/\Lambda_\alpha, dz) \#_\beta (\mathbb{C}/\Lambda_\alpha, dz) \#_\beta \cdots \#_\beta (\mathbb{C}/\Lambda_\alpha, dz)$$

be the genus p translation surface formed by gluing p copies of the same flat torus along a slit of holonomy $(\beta, 1/2)$.

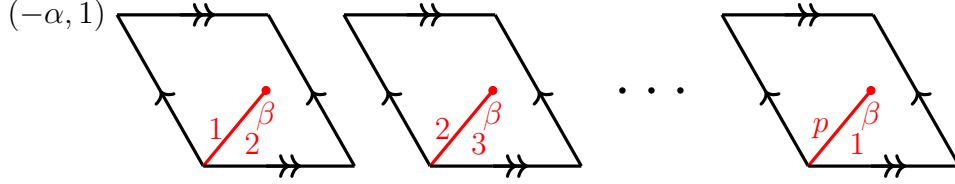
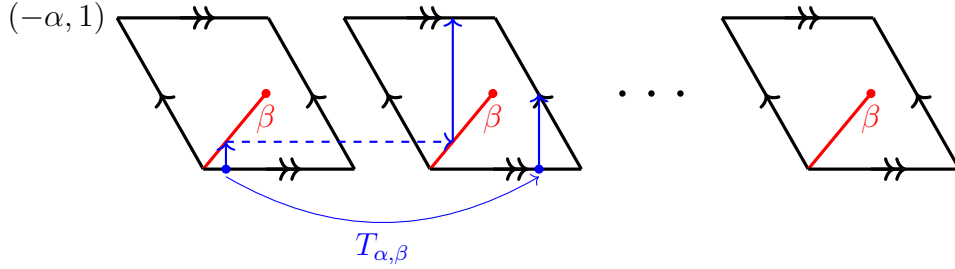


Figure 3: Polygon representation of surface $X_{\alpha,\beta}$

Lemma 1.1.3. $T_{\alpha,\beta}$ is the First return map to the horizontal segment of length p of the vertical flow on $(X_{\alpha,\beta}, \omega)$.

The reader is invited to convince himself by looking at the following picture.



1.1.3 Main result

Recall that any irrational number $u \in (0, 1)$ admits a unique infinite continued fraction expansion $u = [a_1, a_2, \dots]$. Its n -th convergent, denoted by $p_n/q_n = [a_1, \dots, a_n]$, satisfies the well-known best approximation property: for every $0 < q \leq q_n$ and $p \in \mathbb{Z}$,

$$\left| u - \frac{p}{q} \right| \leq \left| u - \frac{p_n}{q_n} \right|.$$

To analyze the location of β with respect to the orbit of the rotation, we employ the Ostrowski numeration system.

Theorem 1.1.4 (Ostrowski expansion). *Let $u \in (0, 1)$ be an irrational number with the continued fraction expansion $[a_1, a_2, \dots]$. Every real number $w \in [0, 1)$ can be uniquely represented in the form*

$$w = \sum_{n=0}^{\infty} b_n |q_n u - p_n|,$$

where the coefficients $(b_n)_{n \geq 0}$ satisfy:

- (i) $0 \leq b_n \leq a_{n+1}$ for all $n \geq 0$;
- (ii) if $b_n = a_{n+1}$ for some $n \geq 0$, then $b_{n+1} = 0$;

(iii) the sequence $(b_n)_{n \geq 1}$ does not terminate with the alternating pattern $b_{N+2k} = a_{N+2k+1}$ and $b_{N+2k+1} = 0$ for all $k \geq 0$.

We are now ready to state the main theorem of this thesis, the case of $p = 2$ is originally due to Veech [Vee69]:

Theorem 1.1.5 (Veech). *Let $\alpha = [a_1, a_2, \dots] \in (0, 1) \setminus \mathbb{Q}$ and let p_n/q_n be its n -th convergent. Suppose $\beta \notin \mathbb{Z}\alpha \pmod{1}$, and let the Ostrowski expansion of β with respect to α be given by*

$$\beta = \sum_{n=0}^{\infty} b_n |q_n \alpha - p_n|.$$

If $p \mid b_n$ for every $n \geq 0$, then the skew-product $T_{\alpha, \beta}$ is non-ergodic with respect to the product measure $\mu \times m$ if

$$\sum_{n=0}^{\infty} b_n q_n |q_n \alpha - p_n| < \infty. \quad (1.1)$$

From the standard theory of continued fractions, the convergence condition (1.1) is known to be equivalent to

$$\sum_{n=0}^{\infty} \frac{b_n}{a_{n+1}} < \infty.$$

From this, one could observe that if α is of bounded type, that is, its continued expansion coefficients a_n form a bounded sequence, condition (1.1) can not be satisfied. In fact, α being of unbounded type is actually necessary for the existence of β to make $T_{\alpha, \beta}$ minimal but not ergodic with respect to $\mu \times m$. A more general version about linear recurrent IETs of this results is shown in [FH24].

Theorem 1.1.6 (Ferenczi, Hubert). *Let $\alpha \in (0, 1) \setminus \mathbb{Q}$ be an irrational number of bounded type. Then for any $\beta \in (0, 1)$, $T_{\alpha, \beta}$ is uniquely ergodic.*

1.2 Further directions

Generic points Suppose α, β are as in the statement of theorem 1.1.5. By a very general observation (see 3.3), there are exactly p ergodic probability measures μ_j , $j \in \{0, 1, \dots, p-1\}$ for $T_{\alpha, \beta}$ and

$$\mu = \frac{1}{p} \sum_{j=0}^{p-1} \mu_j.$$

Question 1. Can we characterize the set of generic points for the ergodic measure μ_j ?

The divisibility condition. Fix an irrational $\alpha = [a_1, a_2, \dots] \in (0, 1) \setminus \mathbb{Q}$ of unbounded type, and let p_n/q_n denote its n -th convergent. Let $K_{\text{NUE}}(\alpha)$ be the set of parameters $\beta \in (0, 1)$ for which the skew-product $T_{\alpha, \beta}$ is minimal but not uniquely ergodic. Using the Ostrowski

expansion $\beta = \sum_{n=0}^{\infty} b_n |q_n \alpha - p_n|$, we define

$$K_1(\alpha) = \left\{ \beta : p \mid b_n \text{ for all } n \geq 0, \sum_{n=0}^{\infty} \frac{b_n}{a_{n+1}} < \infty \right\},$$

$$K_2(\alpha) = \left\{ \beta : p \mid b_n \text{ for all } n \geq 0, \lim_{n \rightarrow \infty} \frac{b_n}{a_{n+1}} = 0 \right\}.$$

Theorem 1.1.5 gives the sufficient condition $K_1(\alpha) \subset K_{\text{NUE}}(\alpha)$. Veech also proved the reverse inclusion $K_{\text{NUE}}(\alpha) \subset K_2(\alpha)$ and asked for an exact description of $K_{\text{NUE}}(\alpha)$ in [Vee69]. Later, Cheung and Eskin [CE07] studied the geometric models via Teichmüller dynamics and sketched a proof of the equality

$$K_{\text{NUE}}(\alpha) = K_1(\alpha).$$

Although this gives a complete arithmetic characterization, the known arguments rely on the geometry of translation surfaces and Teichmüller dynamics. A natural question, and one that suggests a further direction for research, is the following:

Question 2. Can the necessity of the divisibility condition $p \mid b_n$ be understood purely within the language of interval exchange transformations and renormalization? Furthermore, is it possible to give a self-contained proof of $K_{\text{NUE}}(\alpha) = K_1(\alpha)$ using only combinatorial and renormalization techniques?

2 Irrational Rotation and its Renormalization

2.1 Continued Fractions Expansion and Irrational Rotation

We begin by collecting some useful facts about the continued fraction expansion and its connections to irrational rotations.

Continued Fractions Expansion. Let $\alpha \in (0, 1) \setminus \mathbb{Q}$ with the continued fraction expansion $\alpha = [a_1, a_2, \dots]$ and convergents

$$\frac{p_n}{q_n} = \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\dots + \frac{1}{a_n}}}}.$$

Set $p_{-1} = 1, q_{-1} = 0, p_0 = 0, q_0 = 1$.

Theorem 2.1.1. *The following holds.*

(i). *Recursive formulas:*

$$\begin{aligned} p_{n+1} &= a_{n+1}p_n + p_{n-1}; \\ q_{n+1} &= a_{n+1}q_n + q_{n-1}; \\ p_n q_{n+1} - q_n p_{n+1} &= (-1)^{n+1}. \end{aligned}$$

(ii). *Alternating property:*

$$\frac{p_0}{q_0} < \frac{p_2}{q_2} < \dots < \alpha < \dots < \frac{p_3}{q_3} < \frac{p_1}{q_1}.$$

(iii). *Upper and lower bounds: for $n \geq 0$,*

$$\frac{1}{q_n(q_n + q_{n+1})} < \left| \alpha - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}.$$

(iv). *Best approximation property: for any integers p, q with $0 < q < q_{n+1}$ and any $n \geq 0$,*

$$\left| \alpha - \frac{p}{q} \right| \geq \left| \alpha - \frac{p_n}{q_n} \right|.$$

and the equality holds if and only if $p/q = p_n/q_n$.

Proof. See Appendix A. ■

Irrational Rotations. Consider the irrational rotation $R_\alpha : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$, given by $x \mapsto x + \alpha \pmod{1}$. Let d denote the metric on $\mathbb{R}/\mathbb{Z}$ induced by the standard metric on $\mathbb{R}$. We denote the signed discrepancy by $\delta_n := q_n \alpha - p_n$ and its absolute value by $\theta_n := |\delta_n|$ for $n \geq 0$. From the alternating property in Theorem 2.1.1(ii), we have $\delta_n = (-1)^n \theta_n$.

For any $x \in \mathbb{R}/\mathbb{Z}$, we have the estimate

$$\frac{1}{q_n + q_{n+1}} < d(x, R_\alpha^{q_n} x) < \frac{1}{q_{n+1}}$$

from Theorem 2.1.1(iii) and the best approximation property (Theorem 2.1.1(iv)) implies for any integer $0 < q \neq q_n < q_{n+1}$,

$$\theta_n = d(x, R_\alpha^{q_n} x) < d(x, R_\alpha^q x).$$

2.2 The Rokhlin Towers and Induced Dynamics

We now construct the Rokhlin towers for the irrational rotation R_α . Keeping the notation $\delta_n = q_n \alpha - p_n$ and $\theta_n = |\delta_n|$ as above. Recall that by the alternating property, δ_n and δ_{n+1} have opposite signs.

To define the base intervals precisely, let $\pi : \mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z}$ be the canonical projection. For any real numbers a, b , we define the half-open interval $I(a, b)$ as follows: if $a < b$, $I(a, b) = [a, b)$; if $b < a$, $I(a, b) = [b, a)$. We then set the base intervals to be the projections of these intervals from the real line:

$$I_n := \pi(I(0, \delta_n)), \quad n \geq 0.$$

By definition, the lengths of these intervals are $|I_n| = \theta_n$. Moreover, since δ_n and δ_{n+1} possess opposite signs, 0 is an interior point of the union $I^{(n)} := I_n \cup I_{n+1}$ on the circle $\mathbb{R}/\mathbb{Z}$, and in particular, and consequently $I^{(n)}$ is a connected half-open interval (homeomorphic to $[0, 1)$).

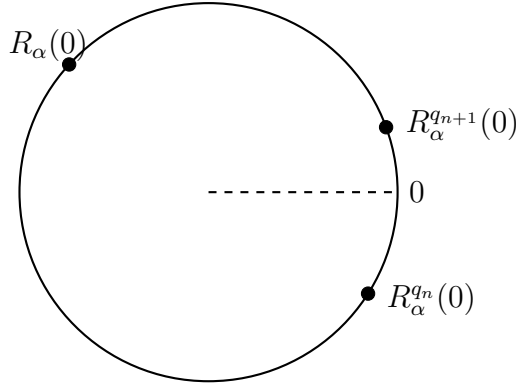


Figure 4: $R_\alpha^{q_n}(0)$ and $R_\alpha^{q_{n+1}}(0)$ are on the different sides of 0, $I^{(n)}$ is the arc with end points $R_\alpha^{q_n}(0)$ and $R_\alpha^{q_{n+1}}(0)$

Let $T^{(n)} : I^{(n)} \rightarrow I^{(n)}$ be the induced map of R_α on $I^{(n)}$, that is,

$$T^{(n)}(x) = R_\alpha^{r_n(x)} x, \quad x \in I^{(n)},$$

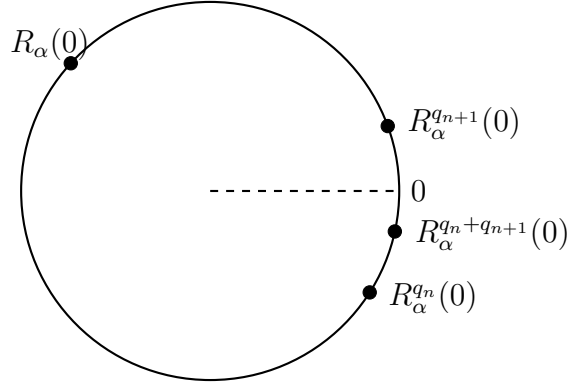
where

$$r_n(x) := \min\{m \geq 1 : R_\alpha^m x \in I^{(n)}\}$$

is the first return time.

Theorem 2.2.1. For $n \geq 0$,

$$T^{(n)}(x) = \begin{cases} R_\alpha^{q_{n+1}} x, & x \in I_n; \\ R_\alpha^{q_n} x, & x \in I_{n+1}. \end{cases}$$

Figure 5: Dynamics of $T^{(n)}$

The figure 5 can help us visualize the dynamics of $T^{(n)}$ defined as in the statement. Recall I_n and I_{n+1} are the arcs with end points $R_\alpha^{q_n}(0), 0$ and $R_\alpha^{q_{n+1}}(0), 0$ respectively. Since θ_n is a decreasing sequence, then $R_\alpha^{q_n+q_{n+1}}(0) \in I^{(n)}$. Therefore,

- $R_\alpha^{q_n}(I_{n+1})$ would be exactly the arc with end points $R_\alpha^{q_n+q_{n+1}}(0)$ and $R_\alpha^{q_n}(0)$
- $R_\alpha^{q_{n+1}}(I_n)$ would be exactly the arc with end points $R_\alpha^{q_n+q_{n+1}}(0)$ and $R_\alpha^{q_{n+1}}(0)$

This shows that $T^{(n)}$ is a well-defined bijective map from $I^{(n)}$ to itself. What remains is to show q_n, q_{n+1} are exactly the first return time of points in I_{n+1} and I_n under iteration of R_α respectively. This is an immediate corollary of the following:

Lemma 2.2.2. *For any $n \geq 0$, the space $\mathbb{R}/\mathbb{Z}$ admits a partition, up to a finite set of endpoints, consisting of two disjoint towers over the bases I_n and I_{n+1} :*

$$\mathbb{R}/\mathbb{Z} = \bigsqcup_{k=0}^{q_{n+1}-1} R_\alpha^k(I_n) \sqcup \bigsqcup_{\ell=0}^{q_n-1} R_\alpha^\ell(I_{n+1}).$$

This decomposition is called the n -th *Kakutani-Rokhlin skyscraper* for R_α . It consists of two Rokhlin towers, denoted by

$$\mathcal{T}_n^{(T)} = \bigsqcup_{k=0}^{q_{n+1}-1} R_\alpha^k(I_n), \quad \mathcal{T}_n^{(S)} = \bigsqcup_{k=0}^{q_n-1} R_\alpha^k(I_{n+1}).$$

Proof. We begin with the disjointness.

- Suppose $R_\alpha^{k_1}(I_n) \cap R_\alpha^{k_2}(I_n) \neq \emptyset$ for integers $0 \leq k_1 < k_2 \leq q_{n+1} - 1$. Then

$$I_n \cap R_\alpha^{k_2-k_1}(I_n) \neq \emptyset.$$

This implies there exists some $x \in I_n$ such that

$$d(x, R_\alpha^{k_2-k_1}x) < |I_n| = \theta_n.$$

By the best approximation property, $k_2 - k_1$ has to be larger than q_{n+1} , which is clearly a contradiction.

- $R_\alpha^{\ell_1}(I_{n+1}) \cap R_\alpha^{\ell_2}(I_{n+1}) = \emptyset$ for integers $0 \leq \ell_1 < \ell_2 \leq q_n - 1$ for the same reason.

- Suppose $R_\alpha^k(I_n) \cap R_\alpha^\ell(I_{n+1}) \neq \emptyset$ for integers $0 \leq k \leq q_{n+1} - 1$ and $0 \leq \ell \leq q_n - 1$. Shifting backward by $\min(k, \ell) \cdot \alpha$ yields two subcases:
 - If $k < \ell$, let $m = \ell - k$. Then $I_n \cap R_\alpha^m(I_{n+1}) \neq \emptyset$ for $0 < m \leq q_n - 1$. This implies there exists some $x \in I_{n+1}$ such that $R_\alpha^m x \in I_n$. Since I_n and I_{n+1} are adjacent on opposite sides of 0, the distance between any two of their points satisfies

$$d(x, R_\alpha^m x) < |I_n| + |I_{n+1}| = \theta_n + \theta_{n+1}.$$

By the best approximation property, for $0 < m < q_n$, the translation distance is bounded below by θ_{n-1} . Since $\theta_{n-1} = a_{n+1}\theta_n + \theta_{n+1} \geq \theta_n + \theta_{n+1}$, $d(x, R_\alpha^m x)$ has to be larger than $\theta_n + \theta_{n+1}$, which is clearly a contradiction.

- If $k \geq \ell$, let $m = k - \ell$. If $m = 0$, $I_n \cap I_{n+1} = \emptyset$ by definition. If $0 < m \leq q_{n+1} - 1$, then $R_\alpha^m(I_n) \cap I_{n+1} \neq \emptyset$. This implies there exists some $x \in I_n$ such that $R_\alpha^m x \in I_{n+1}$. This requires a translation from I_n to I_{n+1} , meaning the displacement $m\alpha \pmod{1}$ must be in the direction of I_{n+1} with a magnitude strictly less than $\theta_n + \theta_{n+1}$. However, by the properties of intermediate fractions, for $0 < m < q_{n+1}$, the closest return to 0 on the I_{n+1} side is exactly the displacement $\delta_{n+1} - \delta_n$. Thus,

$$d(x, R_\alpha^m x) \geq |\delta_{n+1} - \delta_n| = \theta_{n+1} + \theta_n,$$

which is clearly a contradiction.

The total Lebesgue measure of the two families of intervals is

$$q_{n+1}\theta_n + q_n\theta_{n+1}.$$

Using the identity $q_{n+1}p_n - q_n p_{n+1} = (-1)^{n+1}$ from Theorem 2.1.1(i) and the relation $\delta_n = q_n\alpha - p_n$, one computes

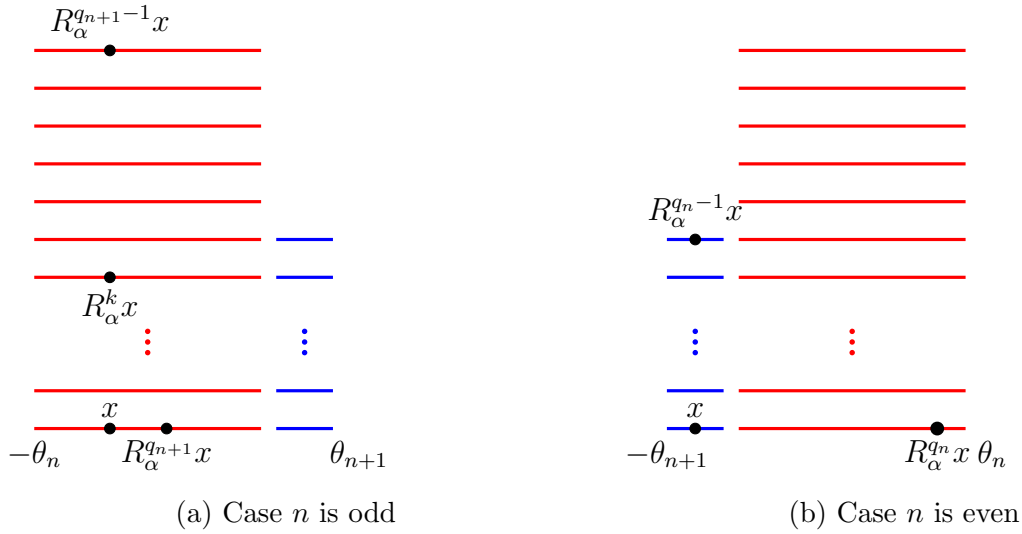
$$q_{n+1}\theta_n + q_n\theta_{n+1} = |q_{n+1}\delta_n| + |q_n\delta_{n+1}| = 1.$$

The disjointness of the levels within each tower follows from the best approximation property: the numbers q_n and q_{n+1} are the first return times to the base intervals, so no intermediate iterate can overlap with the base or with each other. Consequently, the two towers are pairwise disjoint and their union has full measure in $\mathbb{R}/\mathbb{Z}$, yielding the desired partition up to a finite set of boundary points. ■

The partition given in the Lemma 2.2.2 together with Theorem 2.2.1 can significantly help us visualize the dynamics of R_α . The picture below explains where the terminologies 'skyscraper' and 'tower' comes from.

The circle is tiled, up to a finite set of endpoints, by the iterates of the two base intervals I_n and I_{n+1} . These iterates are pairwise disjoint and are arranged into two towers:

$$\mathcal{T}_n^{(T)} = \bigsqcup_{k=0}^{q_{n+1}-1} R_\alpha^k(I_n), \quad \mathcal{T}_n^{(S)} = \bigsqcup_{\ell=0}^{q_n-1} R_\alpha^\ell(I_{n+1}).$$

Figure 6: Rokhlin towers representation for R_α .

- Inside each tower, the rotation R_α simply moves a point one level up.
- The behaviour at the top of the towers is dictated by the induced map $T^{(n)}$. According to Theorem 2.2.1, the top level of the towers is map back to the base by R_α

Thus, the action of R_α on the whole circle can be visualized as a point climbing two parallel stacks; whenever it reaches the top of one stack, it instantaneously jumps to the bottom of the other stack and continues to climb.

This picture also make the rigidity of irrational rotations geometrically evident. The heights of the towers are the denominators q_{n+1} and q_n of the convergents of α , while the widths of the base intervals are the discrepancies θ_n and θ_{n+1} . Because R_α is a rigid translation, the orbit of any point must follow the tower levels in a perfectly regular fashion. The well-known small denominator phenomenon—the fact that $q_n\alpha$ is extremely close to an integer—is reflected in the narrow width of the towers: the smaller the width, the taller the towers must be to cover the circle. As n grows, one obtains a sequence of towers that become increasingly slender and tall, yielding a renormalization scheme that reveals the fine combinatorial structure of the orbit.

2.3 Ostrowski expansion and its Dynamical Interpretation

Theorem 2.3.1 (Ostrowski expansion). *Let $\alpha \in (0, 1) \setminus \mathbb{Q}$ with the continued fraction expansion $\alpha = [a_1, a_2, \dots]$ and convergents p_n/q_n . Let $\theta_n := |q_n\alpha - p_n|$ for $n \geq 0$. Then every $\beta \in [0, 1)$ admits a unique expansion of the form*

$$\beta = \sum_{n=0}^{\infty} b_n \theta_n,$$

where the integer coefficients satisfy:

- (i) $0 \leq b_n \leq a_{n+1}$ for all $n \geq 0$;
- (ii) if $b_n = a_{n+1}$ for some $n \geq 1$, then $b_{n+1} = 0$;
- (iii) the sequence $(b_n)_{n \geq 1}$ does not terminate with the alternating pattern $(a_{n+1}, 0, a_{n+3}, 0, \dots)$. That is, there is no $N \geq 1$ such that $b_{N+2k} = a_{N+2k+1}$ and $b_{N+2k+1} = 0$ for all $k \geq 0$.

Proof. We make a convention by defining $\theta_{-1} := 1$. Recall from the theory of continued fraction we have the following identities:

$$\theta_{n-2} = a_n \theta_{n-1} + \theta_n \quad \text{for all } n \geq 1. \quad (2.1)$$

Existence. We construct $(b_n)_{n \geq 0}$ via a greedy algorithm (see Figure 7 below for illustration). By defining $\beta_{-1} = 0$, we can recursively define the coefficients and partial sums for all $n \geq 0$ uniformly as:

$$b_n = \left\lfloor \frac{\beta - \beta_{n-1}}{\theta_n} \right\rfloor \quad \text{and} \quad \beta_n = \beta_{n-1} + b_n \theta_n.$$

By definition of β_n , we have

$$0 \leq \beta - \beta_n = (\beta - \beta_{n-1}) - b_n \theta_n = \left\{ \frac{\beta - \beta_{n-1}}{\theta_n} \right\} \theta_n < \theta_n$$

for all $n \geq 0$. Since $\theta_n \rightarrow 0$ as $n \rightarrow \infty$, it immediately follows that $\beta_n \rightarrow \beta$, establishing the validity of the expansion.

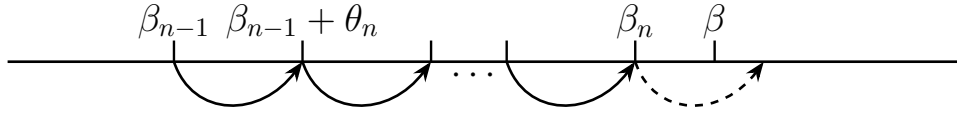


Figure 7: The Greedy algorithm

We now verify inductively that $(b_n)_{n \geq 0}$ satisfies conditions (i)–(iii). For condition (i), applying the strict inequality $\beta - \beta_{n-1} < \theta_{n-1}$ gives

$$b_n = \left\lfloor \frac{\beta - \beta_{n-1}}{\theta_n} \right\rfloor \leq \left\lfloor \frac{\theta_{n-1}}{\theta_n} \right\rfloor = a_{n+1},$$

where the equality holds uniformly for all $n \geq 0$ due to identity (2.1) (and $a_1 = \lfloor 1/\alpha \rfloor = \lfloor \theta_{-1}/\theta_0 \rfloor$).

For condition (ii), suppose that $b_{n-1} = a_n$ for some $n \geq 1$. By the recursive definition, the remaining distance is

$$\beta - \beta_{n-1} = (\beta - \beta_{n-2}) - a_n \theta_{n-1}.$$

Using the inductive bound $\beta - \beta_{n-2} < \theta_{n-2}$ and identity (2.1), we deduce

$$0 \leq \beta - \beta_{n-1} < \theta_{n-2} - a_n \theta_{n-1} = \theta_n.$$

This forces $b_n = \lfloor \frac{\beta - \beta_{n-1}}{\theta_n} \rfloor = 0$, verifying condition (ii).

To verify condition (iii), suppose for the sake of contradiction that the greedy sequence does terminate with the forbidden alternating pattern. That is, there exists some integer $K \geq 1$ such that $b_{K+2m} = a_{K+2m+1}$ and $b_{K+2m+1} = 0$ for all $m \geq 0$.

Under this assumption, the remaining distance after step $M - 1$ would be exactly the sum of this infinite tail:

$$\beta - \beta_{M-1} = \sum_{n=M}^{\infty} b_n \theta_n = \sum_{m=0}^{\infty} a_{M+2m+1} \theta_{M+2m}.$$

By the fundamental identity (2.1), we can rewrite each term as $a_n\theta_{n-1} = \theta_{n-2} - \theta_n$. Substituting this into the sum yields a telescoping series:

$$\begin{aligned}\beta - \beta_{M-1} &= (\theta_{M-1} - \theta_{M+1}) + (\theta_{M+1} - \theta_{M+3}) + (\theta_{M+3} - \theta_{M+5}) + \dots \\ &= \theta_{M-1} - \lim_{m \rightarrow \infty} \theta_{M+2m+1}.\end{aligned}$$

Since $\theta_n \rightarrow 0$, the sum evaluates exactly to θ_{M-1} . Thus, we have $\beta - \beta_{M-1} = \theta_{M-1}$.

However, recall the relationship from the $(M-1)$ -th step of the greedy algorithm:

$$\beta - \beta_{M-2} = b_{M-1}\theta_{M-1} + (\beta - \beta_{M-1}) = b_{M-1}\theta_{M-1} + \theta_{M-1} = (b_{M-1} + 1)\theta_{M-1}.$$

This implies that the ratio at step $K-1$ is exactly an integer:

$$\frac{\beta - \beta_{M-2}}{\theta_{M-1}} = b_{M-1} + 1.$$

Consequently, the greedy choice would dictate $b_{M-1} = \lfloor b_{M-1} + 1 \rfloor = b_{M-1} + 1$, which is a blatant contradiction. Therefore, the greedy algorithm can never produce the forbidden tail, condition (iii) is hence verified.

Uniqueness. For any index $k \geq 1$ and any sequence $(c_n)_{n \geq k}$ satisfying conditions (i), (ii), we claim the strict tail bound:

$$\sum_{n=k}^{\infty} c_n \theta_n < \theta_{k-1}. \quad (2.2)$$

To see this, note $c_k \theta_k \leq a_{k+1} \theta_k = \theta_{k-1} - \theta_{k+1}$. If $c_k = a_{k+1}$, then $c_{k+1} = 0$, and the tail shifts to $k+2$. Iterating this yields the upper bound θ_{k-1} , with equality holding if and only if the forbidden alternating pattern in condition (iii) occurs. Thus, the bound is strictly less than θ_{k-1} .

Suppose β admits two distinct valid expansions $\sum_{n=0}^{\infty} b_n \theta_n$ and $\sum_{n=0}^{\infty} c_n \theta_n$. Applying (2.2) with $k=1$, both infinite tails $\sum_{n=1}^{\infty} b_n \theta_n$ and $\sum_{n=1}^{\infty} c_n \theta_n$ are strictly less than $\theta_0 = \alpha$. Since $\beta \in [0, 1)$, this rigidly determines the first coefficient: $b_0 = c_0 = \lfloor \beta/\alpha \rfloor$.

Let $k \geq 1$ be the smallest index where $b_k \neq c_k$. Assume without loss of generality that $b_k > c_k$. Rearranging the identity between the two tails gives:

$$(b_k - c_k)\theta_k + \sum_{n=k+1}^{\infty} b_n \theta_n = \sum_{n=k+1}^{\infty} c_n \theta_n.$$

Since $b_k - c_k \geq 1$ and the remaining terms are non-negative, the left-hand side is at least θ_k . However, applying (2.2) to the right-hand side yields $\sum_{n=k+1}^{\infty} c_n \theta_n < \theta_k$. This produces the contradiction $\theta_k \leq \text{LHS} = \text{RHS} < \theta_k$. Therefore, the expansion is unique. $\blacksquare$

Remark. In the preceding proof of condition (iii), we actually showed that any sequence ending with such tail is numerically equivalent to a strictly finite expansion obtained by incrementing b_{K-1} by 1 and setting all subsequent coefficients to 0.

Dynamical Interpretation. We now discuss the dynamical meaning of the coefficients b_n in the Ostrowski expansion. Recall that for the irrational rotation R_α , the q_n -th iterate acts as a translation by the closest return distance: $R_\alpha^{q_n}(x) = x + \delta_n \pmod{1}$ for $n \geq 0$, where $\delta_n = q_n\alpha - p_n = (-1)^n\theta_n$.

Following the greedy construction of the sequence (b_n) in our proof, the coefficient b_{n+1} dynamically represents the maximal number of iterations of $R_\alpha^{q_{n+1}}$ (or $R_\alpha^{-q_{n+1}}$, depending on the parity of n to account for the alternating sign of δ_{n+1}) that one can apply to the partial sum β_n before it strictly passes the target point β . To capture this greedy arithmetic process geometrically, we can embed these b_{n+1} translational steps into the Kakutani-Rokhlin tower. Let $\mathcal{T}_{n+1}^{(T)}$ denote the taller tower in the $(n+1)$ -th Kakutani-Rokhlin skyscraper. Notice that

$$\beta_m = \sum_{k=0}^m b_k \theta_k = \sum_{k=0}^m (-1)^k b_k (q_k \alpha - p_k) = \sum_{k=0}^m (-1)^k b_k q_k \alpha \in \mathbb{Z}\alpha \pmod{1}. \quad \forall m \in \mathbb{N}.$$

The dynamical evolution of the interval connecting β_n to β_{n+1} naturally generates a specific subtower.

Definition 2.3.2 (Ostrowski Tower). Let $J_n = [\beta_n, \beta_n + \theta_{n+1})$, if n is odd and $J_n = [\beta_{n+1} - \theta_{n+1}, \beta_{n+1})$ if n is even. We call the **subtower** inside the taller tower of a $(n+1)$ -th Kakutani-Rokhlin skyscraper defined as follows

$$\mathcal{O}_n := \bigsqcup_{k=0}^{(b_{n+1}-1) \cdot q_{n+1}} R_\alpha^k(J_n) \subseteq \mathcal{T}_{n+1}^{(T)},$$

where the n -th Ostrowski tower for R_α .

Notice that the interval $[\beta_n, \beta_{n+1})$ is exactly the disjoint union of the (kq_{n+1}) -th floors of the Ostrowski tower $\mathcal{O}_n$ for $0 \leq k \leq b_{n+1} - 1$.

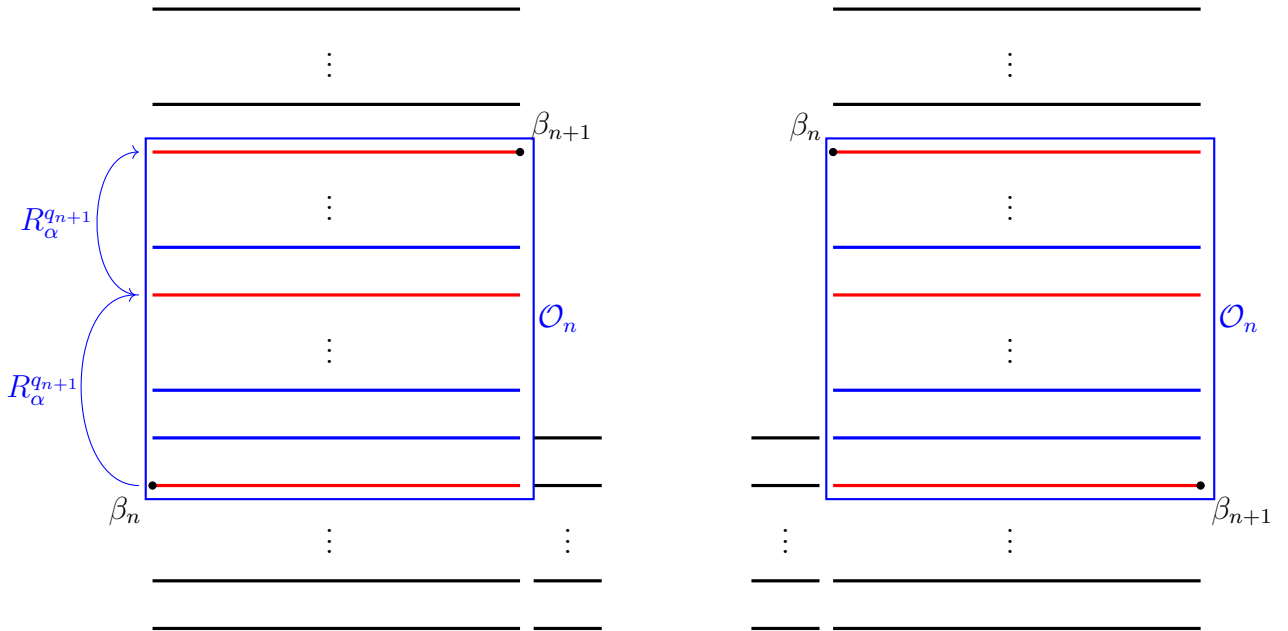


Figure 8: Examples of n -th Ostrowski towers

In Figure 8 we give examples of n -th Ostrowski towers with n being even on the left and n being odd on the right. The subtower colored in blue is the n -th Ostrowski tower $\mathcal{O}_n$ and the floors with red color are parts of the interval $[\beta_n, \beta_{n+1})$. One should be careful that in the case n being odd, β_n should be at the higher floor in the corresponding tower since J_n is in the orbit of $[\beta_n, \beta_n + \theta_n)$ under R_α^{-1} , as we discussed in the first paragraph of this subsection.

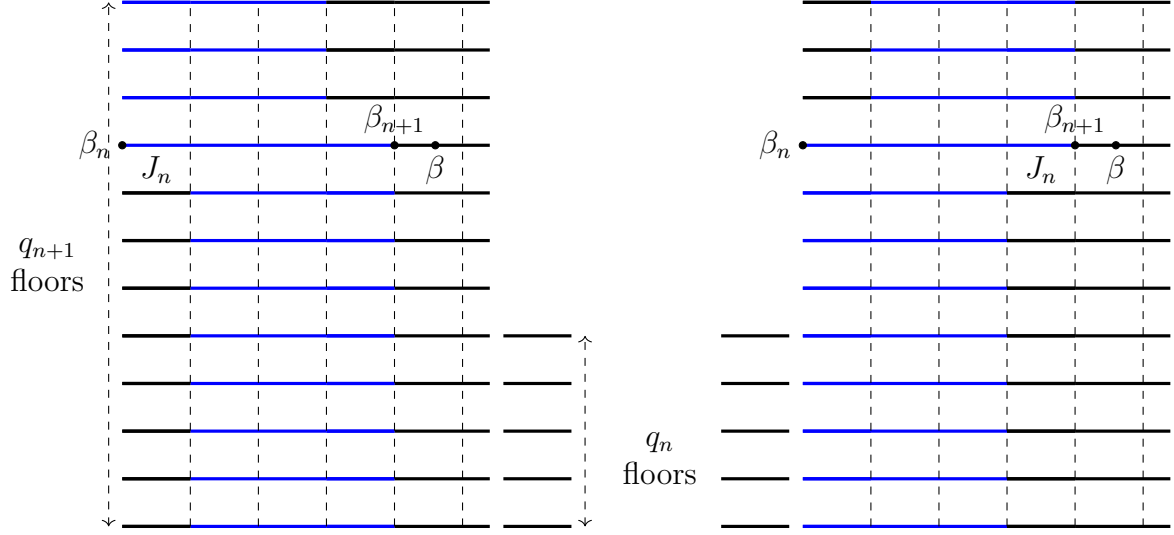


Figure 9: Examples of n -th Ostrowski towers inside $\mathcal{T}_n^{(T)}$

One can also see the n -th Ostrowski tower in $\mathcal{T}_n^{(T)}$, which are the blue parts in Figure 9. Here, the information that we always choose β_{n+1} at the right side of β_n and at the left side of β is kept (i.e. we always approximate β from the left). Notice that after a 'cut and stack', one can go from Figure 9 to Figure 8.

3 Renormalization and Cohomological Equations

In this chapter, we will establish a slightly more general version of Theorem 1.1.5 using the results discussed in Chapter 2.

3.1 Ergodicity vs. Cohomological Equations

Our first step in studying the ergodicity of the group extension is to reduce it to a dynamical problem on the base system, an idea first appeared in Furstenberg's seminal framework [Fur61]. In this section, we achieve this goal via the Fourier analysis on the fiber. Specifically, we use the following criterion:

Proposition 3.1.1. *Let $(X, \mathcal{B}, \mu, T)$ be an invertible, ergodic measure-preserving system, and let $p \geq 2$ be a prime. Consider the skew product*

$$T_\phi : X \times \mathbb{Z}/p\mathbb{Z} \rightarrow X \times \mathbb{Z}/p\mathbb{Z}, \quad (x, j) \mapsto (Tx, j + \phi(x)),$$

where $\phi : X \rightarrow \mathbb{Z}/p\mathbb{Z}$ is a measurable cocycle. Then T_ϕ is ergodic with respect to the product measure $\mu \times m$ (where m is the uniform probability measure on $\mathbb{Z}/p\mathbb{Z}$) if and only if the following functional equation, (usually called cohomological equation in the literature)

$$f \circ T = \phi + f \pmod{p}$$

has no measurable solution $f : X \rightarrow \mathbb{Z}/p\mathbb{Z}$.

Proof. Denote $G = \mathbb{Z}/p\mathbb{Z}$ and let $\chi_k(j) = e^{2\pi i k j / p}$ be the characters of G , for $k = 0, 1, \dots, p-1$. For any $F \in L^2(X \times G)$, we can compute its Fourier coefficients with respect to the fiber:

$$\hat{F}_k(x) = \sum_{j \in G} F(x, j) \overline{\chi_k(j)} = \sum_{j \in G} F(x, j) e^{-2\pi i k j / p}.$$

This yields the orthogonal expansion

$$F(x, j) = \frac{1}{p} \sum_{k=0}^{p-1} \hat{F}_k(x) \chi_k(j).$$

By a direct computation, the invariance condition $F \circ T_\phi = F$ is equivalent to the system of equations:

$$\hat{F}_k(Tx) = \chi_k(-\phi(x)) \hat{F}_k(x) \quad \mu\text{-a.e.} \quad (3.1)$$

for each $k \in \{0, \dots, p-1\}$.

Measurable solution implies non-ergodicity. Suppose there exists a measurable function $f : X \rightarrow G$ such that $f(Tx) = \phi(x) + f(x) \pmod{p}$. Consider the graph of f :

$$\Gamma_f := \{(x, f(x)) : x \in X\} \subseteq X \times \mathbb{Z}/p\mathbb{Z}.$$

Then

$$T_\phi(\Gamma_f) = \{(Tx, f(x) + \phi(x)) : x \in X\} = \{(Tx, f(Tx)) : x \in X\} \xlongequal{T \text{ is invertible}} \Gamma_f.$$

Since $\mu \times m(\Gamma_f) = \frac{1}{p}$, and Γ_f is T_ϕ invariant, we conclude that T_ϕ is not ergodic.

Non-ergodicity implies a measurable solution. Suppose T_ϕ is not ergodic. Then there exists a non-trivial T_ϕ -invariant function F . From (3.1), there is some $k \neq 0$ such that $\hat{F}_k$ is not identically zero; otherwise, F would depend only on x and be constant by the ergodicity of T .

Taking the absolute value of the both sides in equation

$$\hat{F}(Tx) = e^{-\frac{2\pi i k \phi(x)}{p}} \hat{F}(x),$$

we see that $|\hat{F}|$ is T -invariant. By the ergodicity of T , $|\hat{F}|$ is a positive constant almost everywhere. Let $H = \hat{F}/|\hat{F}|$; then H takes values in the unit circle and satisfies the exact same equation:

$$H(Tx) = e^{-\frac{2\pi i k \phi(x)}{p}} H(x). \quad (3.2)$$

Raising both sides to the p -th power, and noting that $e^{-2\pi i k \phi(x)} = 1$, we obtain $H^p(Tx) = H^p(x)$. Thus, H^p is T -invariant and, by ergodicity, must be constant almost everywhere. This implies that h takes values in a rotated set of p -th roots of unity. We can therefore write $H(x) = e^{2\pi i h(x)/p}$ for some measurable function $h : X \rightarrow \mathbb{Z}/p\mathbb{Z}$.

Substituting this expression back into the equation (3.2) yields:

$$e^{2\pi i h(Tx)/p} = e^{2\pi i (h(x) - k\phi(x))/p} \implies h(Tx) = h(x) - k\phi(x) \pmod{p}.$$

Crucially, since p is prime and $k \not\equiv 0 \pmod{p}$, k is invertible modulo p . Let ℓ be an integer such that $\ell k = 1 \pmod{p}$. Multiplying the congruence by $-\ell$ gives:

$$-\ell h(Tx) = -\ell h(x) + \phi(x) \pmod{p}.$$

Setting $f = -\ell h \pmod{p}$, we arrive at the exact cohomological equation:

$$f \circ T = \phi + f \pmod{p}.$$

Therefore, we have established that T_ϕ is ergodic if and only if the equation $f \circ T = \phi + f \pmod{p}$ admits no measurable solution. ■

Remark. For readers less familiar with Fourier analysis over finite groups, it is highly instructive to first verify the case $p = 2$. The Fourier expansion then reduces to the classical orthogonal decomposition of a function $F(x, j)$ into its even and odd parts with respect to the fiber:

$$F(x, j) = \frac{F(x, 0) + F(x, 1)}{2} + (-1)^j \frac{F(x, 0) - F(x, 1)}{2}.$$

The proof above is the natural algebraic generalization of this elementary procedure.

For the rest of this chapter, we fix $\alpha \in (0, 1) \setminus \mathbb{Q}$ with the continued fraction expansion $[a_1, a_2, \dots]$ and convergents p_n/q_n for $n \geq 0$. Let $\beta \in (0, 1) \setminus \mathbb{Z}\alpha$, whose Ostrowski expansion with respect to α is given by

$$\beta = \sum_{n=0}^{\infty} b_n \theta_n.$$

such that p divides b_n . Let $R_\alpha : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$, $x \mapsto x + \alpha$ be the irrational rotation, and μ be the Lebesgue measure on $\mathbb{R}/\mathbb{Z}$. We consider the skew product given by the step cocycle

$$T_{\alpha,\beta} := \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \longrightarrow \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}, \quad (x, j) \longmapsto (R_\alpha x, j + \mathbf{1}_{[0,\beta)}(x)).$$

From the previous proposition, we have

Lemma 3.1.2. *$T_{\alpha,\beta}$ is ergodic with respect to $\mu \times m$ if and only if the functional equation*

$$f \circ R_\alpha = \mathbf{1}_{[0,\beta)} + f \pmod{p}$$

has no measurable solution $f : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$.

3.2 Non-ergodic Approximations

For each $n \geq 0$, let $\beta_n = \sum_{k=0}^n b_k \theta_k$ be the n -th truncation of the Ostrowski expansion of β . We define the approximating transformation

$$T_n : \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \longrightarrow \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}, \quad (x, j) \longmapsto (R_\alpha x, j + \mathbf{1}_{[0,\beta_n)}(x)),$$

with the convention that $T_{-1}(x, j) = (x + \alpha, j)$. Since $\beta_n \rightarrow \beta$ as $n \rightarrow \infty$, the step cocycles $\mathbf{1}_{[0,\beta_n)}$ converge to $\mathbf{1}_{[0,\beta)}$ in $L^1(\mu)$. In other words, the sequence of transformations T_n serves as a natural measure-theoretic approximation to the target system T .

The goal of this section is to show that that, under the arithmetic assumption that p divides b_n for all $n \geq 0$, the transformation T_n is strictly non-ergodic for every $n \geq 0$. By Proposition 3.1.1, the non-ergodicity of T_n with respect to the product measure $\mu \times m$ is equivalent to the existence of a measurable function $f : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$ satisfying the following cohomological equation:

$$f \circ R_\alpha = \mathbf{1}_{[0,\beta_n)} + f \pmod{p} \quad \mu\text{-a.e.} \quad (\text{n})$$

The following theorem explicitly constructs the measurable solutions to these approximating equations, which will be useful in next section.

Theorem 3.2.1. *For every $n \geq -1$, the transformation T_n is not ergodic with respect to the product measure $\mu \times m$. Specifically, if $f_n : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$ is a measurable solution to the equation (n), then $f_n + 1$ is also a solution to (n). Moreover, if $b_{n+1} = 0$, define $f_{n+1} = f_n$ and otherwise define f_{n+1} as follows:*

$$f_{n+1}(x) := \begin{cases} f_n(x) + (k+1) \pmod{p}, & x \in \mathcal{Z}_{n,k} \\ f_n(x), & \text{otherwise,} \end{cases}$$

where

$$\mathcal{Z}_{n,k} = \bigcup_{\ell=1}^{q_{n+1}} R_\alpha^{kq_{n+1}+\ell}(J_n), \quad k \in \{0, 1, \dots, b_{n+1} - 1\}.$$

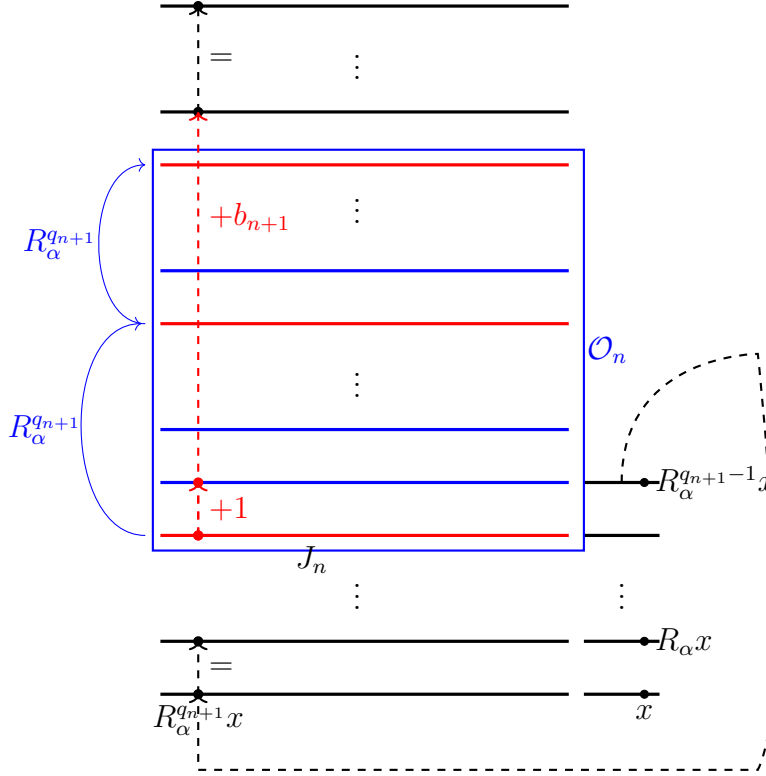
$J_n = [\beta_n, \beta_n + \theta_{n+1})$ if n is even, $J_n = [\beta_{n+1} - \theta_{n+1}, \beta_{n+1})$ if N is odd, f_{n+1} defined as above is a measurable solution to equation (n+1).

First, observe that since $\mathbf{1}_{[0,\beta_{n+1})} = \mathbf{1}_{[0,\beta_n)} + \mathbf{1}_{[\beta_n, \beta_{n+1})}$, to solve equation (n+1), it suffices to solve

$$g \circ R_\alpha = \mathbf{1}_{[\beta_n, \beta_{n+1})} + g.$$

Intuitively, the dynamical meaning of the formula for f_{n+1} comes from the following procedure (also see Figure 10) of solving the above equation for g .

- For $x \in \mathcal{T}_{n+1}^{(S)}$, set $g(x) = 0$.
- The top floor of $\mathcal{T}_{n+1}^{(S)}$ will move to the base of $\mathcal{T}_{n+1}^{(T)}$ after applying R_α . Set $g(x) = 0$ for $x \in I_{n+1}$.
- We define g floor by floor. The value on the next floor is the same as on the current one, unless the current one is $R^{kq_{n+1}}(J_n)$, in that case, we set $g(R_\alpha x) = g(x) + 1$.
- When exists $\mathcal{O}_n$, we add in total b_{n+1} times 1. Since $p \mid b_{n+1}$, the value of g in mod p sense is the same as before entering $\mathcal{O}_n$.

Figure 10: Solving $g \circ R_\alpha = \mathbf{1}_{[\beta_n, \beta_{n+1})} + g$

Proof of Theorem 3.2.1. It is clear that T_{-1} is not ergodic and $f_{-1} \equiv 1$ is a solution to the equation (-1) .

Suppose for $n \geq -1$ we have T_n is not ergodic with respect to the product measure $\mu \times m$ and f_n is a measurable solution to the equation (n). Our goal is to verify that f_{n+1} defined as in the statement is indeed a solution to the equation (n+1). If $b_{n+1} = 0$, then $f_{n+1} = f_n$ is obviously a solution to (n+1). If $b_{n+1} \neq 0$, since $\mathbf{1}_{[0, \beta_{n+1})} = \mathbf{1}_{[0, \beta_n)} + \mathbf{1}_{[\beta_n, \beta_{n+1})}$, it then suffices to show

$$g_n(x) := f_{n+1}(x) - f_n(x) = \begin{cases} k+1 \pmod{p}, & x \in \mathcal{Z}_{n,k}, \\ 0, & \text{otherwise.} \end{cases}$$

satisfies

$$g_n \circ R_\alpha = \mathbf{1}_{[\beta_n, \beta_{n+1})} + g_n \pmod{p}. \quad (3.3)$$

By the properties of the Ostrowski expansion, the disjoint union of the translates $R_\alpha^{kq_{n+1}}(J_n)$ for integers $0 \leq k < b_{n+1}$ exactly partitions the interval $[\beta_n, \beta_{n+1})$ up to a set of μ -measure zero,

that is, $[\beta_n, \beta_{n+1}) = \bigsqcup_{k=0}^{b_{n+1}-1} R_\alpha^{kq_{n+1}}(J_n)$. Recall we have defined the n -th Ostrowski towers for R_α :

$$\mathcal{O}_n := \bigsqcup_{k=0}^{b_{n+1}-1} \bigsqcup_{\ell=0}^{q_{n+1}-1} R_\alpha^{kq_{n+1}+\ell}(J_n) \subset \mathcal{T}_{n+1}^{(T)}.$$

It is a subtower of the taller tower from the $(n+1)$ -th Kakutani-Rokhlin skyscraper. Then $[\beta_n, \beta_{n+1})$ are exactly the disjoint union of the kq_{n+1} -th floors of the $\mathcal{O}_{n+1}$. Define

$$\mathcal{O}_{n,k} := \bigsqcup_{\ell=0}^{q_{n+1}-1} R_\alpha^{kq_{n+1}+\ell}(J_n), \quad k \in \{0, 1, \dots, b_{n+1}-1\}$$

Notice that

$$\mathcal{Z}_{n,k} = R_\alpha(\mathcal{O}_{n,k}), \quad \text{and} \quad \mathcal{Z}_{n,k} \triangle \mathcal{O}_{n,k} = R_\alpha^{kq_{n+1}}(J_n) \sqcup R_\alpha^{(k+1)q_{n+1}}.$$

then we have

$$\begin{aligned} g_n \circ R_\alpha - g_n &= \sum_{k=0}^{b_{n+1}-1} (k+1) \cdot \mathbf{1}_{\mathcal{O}_{n,k}} - \sum_{k=0}^{b_{n+1}-1} (k+1) \cdot \mathbf{1}_{\mathcal{Z}_{n,k}} \\ &= \sum_{k=0}^{b_{n+1}-1} (k+1) \cdot (\mathbf{1}_{\mathcal{O}_{n,k}} - \mathbf{1}_{\mathcal{Z}_{n,k}}) \\ &= \sum_{k=0}^{b_{n+1}-1} (k+1) \cdot \left(\mathbf{1}_{R_\alpha^{kq_{n+1}}(J_n)} - \mathbf{1}_{R_\alpha^{(k+1)q_{n+1}}(J_n)} \right) \\ &= \sum_{k=0}^{b_{n+1}-1} \mathbf{1}_{R_\alpha^{kq_{n+1}}(J_n)} - b_{n+1} \cdot \mathbf{1}_{R_\alpha^{b_{n+1}q_{n+1}}(J_n)} \end{aligned}$$

Since p divides b_{n+1} and $[\beta_n, \beta_{n+1}) = \bigsqcup_{k=0}^{b_{n+1}-1} R_\alpha^{kq_{n+1}}(J_n)$, in $(\bmod p)$ sense, we have

$$g_n \circ R_\alpha - g_n = \sum_{k=0}^{b_{n+1}-1} \mathbf{1}_{R_\alpha^{kq_{n+1}}(J_n)} = \mathbf{1}_{[\beta_n, \beta_{n+1})} \pmod{p}.$$

The proof is now complete. ■

This explicit construction allows us to effortlessly quantify the discrepancy between successive solutions. The following lemma, which provides a crucial error estimate, is an immediate corollary of Theorem 3.2.1.

Lemma 3.2.2. *Let $f_{-1} \equiv 0$, and let $(f_n)_{n \geq 0}$ be the sequence of solutions to the equations (n) constructed as in Theorem 3.2.1. Define the discrepancy set as*

$$\mathcal{D}_n = \{x \in \mathbb{R}/\mathbb{Z} : f_n(x) \neq f_{n+1}(x)\}.$$

Then its Lebesgue measure is given exactly by

$$\mu(\mathcal{D}_n) = \frac{p-1}{p} b_{n+1} q_{n+1} \theta_{n+1}.$$

We are now ready to prove

Theorem 3.2.3. *If*

$$\sum_{n=0}^{\infty} b_n q_n \theta_n < \infty,$$

then $T_{\alpha,\beta}$ is not ergodic with respect to $\mu \times m$.

Proof. Recall from Lemma 3.2.2 that the discrepancy sets $\mathcal{D}_n = \{x \in \mathbb{R}/\mathbb{Z} : f_n(x) \neq f_{n+1}(x)\}$ have measure $\mu(\mathcal{D}_n) = \frac{p-1}{p} b_{n+1} q_{n+1} \theta_{n+1}$. Therefore, the assumption directly guarantees that the sum of the measures is finite:

$$\sum_{n=0}^{\infty} \mu(\mathcal{D}_n) < \infty.$$

By the Borel-Cantelli lemma, the set of points belonging to infinitely many $\mathcal{D}_n$ has Lebesgue measure zero. That is,

$$\mu\left(\limsup_{n \rightarrow \infty} \mathcal{D}_n\right) = \mu\left(\bigcap_{n=0}^{\infty} \bigcup_{k=n}^{\infty} \mathcal{D}_k\right) = 0.$$

Equivalently, for μ -almost every $x \in \mathbb{R}/\mathbb{Z}$, the point x belongs to only finitely many sets $\mathcal{D}_n$. It follows that for μ -almost every x the sequence of values $(f_n(x))_{n \geq -1}$ changes only finitely many times. Thus, the sequence is eventually constant pointwise, which implies that $(f_n)_{n \geq -1}$ converges μ -almost everywhere to a well-defined measurable limit function $f : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$.

Taking the pointwise limit as $n \rightarrow \infty$ on both sides of the approximating cohomological equations

$$f_n \circ R_\alpha - f_n = \mathbf{1}_{[0, \beta_n)} \pmod{p},$$

we obtain the exact cohomological equation

$$f \circ R_\alpha - f = \mathbf{1}_{[0, \beta)} \pmod{p}.$$

By Proposition 3.1.1, the existence of such a measurable solution f concludes the proof. $\blacksquare$

3.3 More on Dynamics of $T_{\alpha,\beta}$

Having established that $T_{\alpha,\beta}$ is not ergodic with respect to the product measure $\mu \times m$ under the arithmetic condition that p divides b_n for all $n \geq 0$, the following general observation can actually allow us completely describe its ergodic components.

Proposition 3.3.1. *Let (X, T, μ) be a uniquely ergodic system, and let p be a prime number. Suppose $\phi : X \rightarrow \mathbb{Z}/p\mathbb{Z}$ is a measurable function, and consider the skew-product extension*

$$T_\phi : X \times \mathbb{Z}/p\mathbb{Z} \longrightarrow X \times \mathbb{Z}/p\mathbb{Z}, \quad (x, \ell) \longmapsto (Tx, \ell + \phi(x)).$$

If T_ϕ is not ergodic with respect to the product measure $\mu \times m$ (where m is the Haar measure on $\mathbb{Z}/p\mathbb{Z}$), then T_ϕ has exactly p ergodic measures.

Proof. Assume that T_ϕ is not ergodic with respect to the product measure $\mu \times m$. By Proposition 3.1.1, the non-ergodicity of the prime-order extension implies that ϕ is a measurable coboundary. That is, there exists a measurable function $f : X \rightarrow \mathbb{Z}/p\mathbb{Z}$ such that for μ -almost every $x \in X$, we have:

$$\phi(x) = f(Tx) - f(x) \pmod{p}$$

We use this coboundary relation to construct a measurable isomorphism between T_ϕ and the trivial product system. Define the map $\Phi : X \times \mathbb{Z}/p\mathbb{Z} \rightarrow X \times \mathbb{Z}/p\mathbb{Z}$ by:

$$\Phi(x, \ell) = (x, \ell - f(x))$$

Clearly, Φ is a measurable bijection that preserves the product measure class. We claim that the following diagram commutes:

$$\begin{array}{ccc} X \times \mathbb{Z}/p\mathbb{Z} & \xrightarrow{T_\phi} & X \times \mathbb{Z}/p\mathbb{Z} \\ \Phi \downarrow & & \downarrow \Phi \\ X \times \mathbb{Z}/p\mathbb{Z} & \xrightarrow{T \times \text{id}} & X \times \mathbb{Z}/p\mathbb{Z} \end{array}$$

Since

$$\Phi(T_\phi(x, \ell)) = \Phi(Tx, \ell + \phi(x)) = (Tx, \ell + \phi(x) - f(Tx)),$$

Substituting the coboundary equation $\phi(x) = f(Tx) - f(x) \pmod{p}$ yields:

$$\Phi(T_\phi(x, \ell)) = (Tx, \ell + f(Tx) - f(x) - f(Tx)) = (Tx, \ell - f(x)).$$

Conversely, applying $T \times \text{id}$ to the image of Φ gives:

$$(T \times \text{id})(\Phi(x, \ell)) = (T \times \text{id})(x, \ell - f(x)) = (Tx, \ell - f(x))$$

Thus, $\Phi \circ T_\phi = (T \times \text{id}) \circ \Phi$, establishing the measurable conjugacy. Consequently, it suffices to classify the ergodic measures of the trivial system $(X \times \mathbb{Z}/p\mathbb{Z}, T \times \text{id})$ and pull them back via Φ .

Let $\tilde{\nu}$ be an arbitrary $(T \times \text{id})$ -invariant probability measure on $X \times \mathbb{Z}/p\mathbb{Z}$, and let $\pi_X : X \times \mathbb{Z}/p\mathbb{Z} \rightarrow X$ denote the canonical projection. The pushforward measure $(\pi_X)_* \tilde{\nu}$ is necessarily T -invariant. By the hypothesis that (X, T) is uniquely ergodic, μ is its only invariant probability measure, so $(\pi_X)_* \tilde{\nu} = \mu$.

By Rokhlin's disintegration theorem, we can disintegrate $\tilde{\nu}$ over μ :

$$\tilde{\nu} = \int_X (\delta_x \times \eta_x) d\mu(x)$$

where $x \mapsto \eta_x$ is a measurable family of probability measures on $\mathbb{Z}/p\mathbb{Z}$. The $(T \times \text{id})$ -invariance of $\tilde{\nu}$ implies that $\eta_{Tx} = \eta_x$ for μ -almost every $x \in X$. Since (X, T, μ) is uniquely ergodic, any T -invariant measurable function is essentially constant. This forces the map $x \mapsto \eta_x$ to be constant μ -a.e., meaning there exists a single probability measure η on $\mathbb{Z}/p\mathbb{Z}$ such that:

$$\tilde{\nu} = \mu \times \eta$$

We have shown that any invariant measure for $T \times \text{id}$ is of the form $\mu \times \eta$. The ergodic measures of a dynamical system correspond to the extreme points of the simplex of its invariant probability measures. Since μ is fixed and ergodic, $\mu \times \eta$ is an extreme point if and only if η is an extreme point in the space of probability measures on $\mathbb{Z}/p\mathbb{Z}$. The extreme points on the finite set $\mathbb{Z}/p\mathbb{Z}$ are precisely the p Dirac measures δ_j for $j \in \{0, 1, \dots, p-1\}$.

Therefore, the trivial system $T \times \text{id}$ possesses exactly p ergodic measures, given by:

$$\tilde{\nu}_j = \mu \times \delta_j, \quad j \in \{0, 1, \dots, p-1\}$$

Finally, we pull these measures back to the original skew-product system T_ϕ using the inverse isomorphism $\Phi^{-1}(x, \ell) = (x, \ell + f(x) \pmod{p})$. This yields exactly p ergodic measures for T_ϕ , explicitly given by:

$$\nu_j = (\Phi^{-1})_* \tilde{\nu}_j = \int_X \delta_{(x, j+f(x))} d\mu(x), \quad j \in \{0, 1, \dots, p-1\}$$

These p measures are mutually singular and are supported on the disjoint invariant graphs defined by the sections $x \mapsto j + f(x)$. This completes the proof. $\blacksquare$

Apply this to our setting, we have

Corollary 3.3.2. *Let $\alpha = [a_1, a_2, \dots] \in (0, 1) \setminus \mathbb{Q}$ and let p_n/q_n be its n -th convergent. Suppose $\beta \notin \mathbb{Z}\alpha \pmod{1}$, and let the Ostrowski expansion of β with respect to α be given by*

$$\beta = \sum_{n=0}^{\infty} b_n \theta_n.$$

If $p \mid b_n$ for every $n \geq 1$, and

$$\sum_{n=0}^{\infty} b_n q_n \theta_n < \infty.$$

then the skew-product $T_{\alpha, \beta}$ has exactly p ergodic measures.

Basically, this tells us that if α, β satisfy the condition to make $T_{\alpha, \beta}$ non-ergodic with respect to $\mu \times m$, the dynamics of $T_{\alpha, \beta}$ in measurable sense is just p independent irrational rotations on the graphs of its p solutions to the cohomological equations.

A Appendix: Continued fraction expansion

We collect some basic facts of the continued fraction expansion here. All the material here can be found in [Khi64]

Lemma A.1. *Fix a sequence $(a_n)_{n \geq 0} \in \mathbb{N}_0 \times \mathbb{N}^{\mathbb{N}}$. The rational numbers*

$$\frac{p_n}{q_n} = [a_0; a_1, a_2, \dots, a_n] := a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\dots + \frac{1}{a_n}}}}, \quad n \geq 0,$$

with $p_n, q_n \geq 1$ and $\gcd(p_n, q_n) = 1$, can be computed recursively via the matrix relation:

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & 1 \\ 1 & 0 \end{pmatrix}, \quad n \geq 0.$$

We set the initial conditions:

$$p_{-1} = 1, \quad q_{-1} = 0, \quad p_0 = a_0, \quad q_0 = 1.$$

Proof. We proceed by induction on n . The base case $n = 0$ is immediate from the initial conditions.

Assume the result holds for all $n \leq k$. Consider the continued fraction

$$\frac{x}{y} = [a_1; a_2, \dots, a_{k+1}], \quad \frac{x'}{y'} = [a_1; a_2, \dots, a_k].$$

By the inductive hypothesis, we have:

$$\begin{pmatrix} x & x' \\ y & y' \end{pmatrix} = \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_{k+1} & 1 \\ 1 & 0 \end{pmatrix}.$$

Multiplying both sides on the left by $\begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix}$, we obtain:

$$\begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x & x' \\ y & y' \end{pmatrix} = \begin{pmatrix} a_0x + y & a_0x' + y' \\ x & x' \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_{k+1} & 1 \\ 1 & 0 \end{pmatrix}.$$

Note that

$$\frac{a_0x + y}{x} = a_0 + \frac{y}{x} = [a_0; a_1, \dots, a_{k+1}] = \frac{p_{k+1}}{q_{k+1}},$$

which completes the inductive step. ■

Corollary A.2. *The sequences (p_n) and (q_n) satisfy the recurrence relations:*

$$\begin{aligned} p_{n+1} &= a_{n+1}p_n + p_{n-1}, \\ q_{n+1} &= a_{n+1}q_n + q_{n-1}, \quad n \geq 0, \end{aligned}$$

with initial conditions:

$$p_0 = a_0, \quad q_0 = 1, \quad p_{-1} = 1, \quad q_{-1} = 0.$$

Since $a_n \geq 1$ for $n \geq 1$, we have:

$$1 = q_0 \leq q_1 < q_2 < \cdots, \quad q_n \geq 2^{(n-3)/2}, \quad (\text{A.1})$$

and

$$a_0 = p_0 \leq p_1 < p_2 < \cdots, \quad p_n \geq 2^{(n-2)/2}.$$

Taking determinants in the matrix relation yields:

$$p_{n+1}q_n - p_nq_{n+1} = (-1)^{n+2} = (-1)^n.$$

Hence,

$$\frac{p_{n+1}}{q_{n+1}} = \frac{p_n}{q_n} + \frac{(-1)^n}{q_nq_{n+1}},$$

and by telescoping,

$$\frac{p_n}{q_n} = a_0 + \sum_{k=0}^{n-1} \frac{(-1)^k}{q_kq_{k+1}}.$$

Corollary A.3. *Let $u := [a_0; a_1, a_2, \dots] = \lim_{n \rightarrow \infty} [a_0; a_1, \dots, a_n]$. Then the limit*

$$u = \lim_{n \rightarrow \infty} \frac{p_n}{q_n} = a_0 + \sum_{k=0}^{\infty} \frac{(-1)^k}{q_kq_{k+1}}$$

always exists, and we have the following alternating inequalities:

$$\frac{p_0}{q_0} < \frac{p_2}{q_2} < \cdots < u < \cdots < \frac{p_3}{q_3} < \frac{p_1}{q_1}.$$

The expression $[a_0; a_1, a_2, \dots]$ is called the continued fraction expansion of u . The continued fraction expansion of any irrational number $u \in \mathbb{R} \setminus \mathbb{Q}$ is unique.

$\{\frac{p_n}{q_n}\}$ converges to u very quickly. Indeed, the error in approximating u by $\frac{p_n}{q_n}$ satisfies:

$$\left| u - \frac{p_n}{q_n} \right| = \left| \sum_{k=n}^{\infty} \frac{(-1)^k}{q_kq_{k+1}} \right| \leq \frac{1}{q_nq_{n+1}} \leq \frac{1}{a_{n+1}q_n^2}.$$

Define $\|u\| := \min_{m \in \mathbb{Z}} |u - m|$. Then the above inequality implies that there exist integers $q_n \rightarrow \infty$ such that

$$q_n \|q_n u\| < \frac{q_n}{q_{n+1}} < \frac{1}{a_{n+1}}.$$

The convergents of the continued fraction expansion are not merely good rational approximants; they are the best possible in the following precise sense.

Proposition A.4 (Optimal Rational Approximation). *Let $u = [a_0; a_1, a_2, \dots] \in \mathbb{R} \setminus \mathbb{Q}$, and let $n \geq 1$. For any integers p, q with $0 < q < q_{n+1}$,*

$$|qu - p| \geq |q_n u - p_n|,$$

where the equality holds if and only if $p/q = p_n/q_n$.

Proof. From Corollary A.3 we know that u lies strictly between $\frac{p_n}{q_n}$ and $\frac{p_{n+1}}{q_{n+1}}$. Consequently, the two quantities

$$\delta_n := q_n u - p_n, \quad \delta_{n+1} := q_{n+1} u - p_{n+1}$$

are non-zero and have opposite signs. Moreover, by Lemma A.1 the matrix

$$M = \begin{pmatrix} p_n & p_{n+1} \\ q_n & q_{n+1} \end{pmatrix}$$

has determinant $p_n q_{n+1} - p_{n+1} q_n = -(-1)^n = \pm 1$. Thus, the matrix M is invertible over $\mathbb{Z}$, and its column vectors form a basis for the integer lattice $\mathbb{Z}^2$. For any given integers p, q , there exist unique integers $x, y \in \mathbb{Z}$ such that

$$\begin{pmatrix} p \\ q \end{pmatrix} = M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x p_n + y p_{n+1} \\ x q_n + y q_{n+1} \end{pmatrix}.$$

Using this linear combination, we can express the approximation error as:

$$qu - p = x(q_n u - p_n) + y(q_{n+1} u - p_{n+1}) = x\delta_n + y\delta_{n+1}.$$

We now evaluate the constraint $0 < q < q_{n+1}$, which translates to $0 < xq_n + yq_{n+1} < q_{n+1}$. We partition our analysis into two cases based on y :

Case 1: $y \neq 0$. If $y \geq 1$, then $xq_n < (1 - y)q_{n+1} \leq 0$. Since $q_n > 0$, this forces $x < 0$. If $y \leq -1$, then $xq_n > -yq_{n+1} \geq q_{n+1} > 0$, which forces $x > 0$. In both scenarios, x and y are non-zero integers with strictly opposite signs. Because δ_n and δ_{n+1} inherently carry opposite signs, the products $x\delta_n$ and $y\delta_{n+1}$ must share the *same* sign. Consequently, the absolute value of their sum is the sum of their absolute values:

$$|qu - p| = |x\delta_n + y\delta_{n+1}| = |x||\delta_n| + |y||\delta_{n+1}|.$$

Given that $x, y \in \mathbb{Z} \setminus \{0\}$, we have $|x| \geq 1$ and $|y| \geq 1$, yielding:

$$|qu - p| \geq |\delta_n| + |\delta_{n+1}| > |\delta_n|.$$

Thus, the strict inequality holds.

Case 2: $y = 0$. Here, $q = xq_n$. The condition $0 < q < q_{n+1}$ ensures that $x \geq 1$. The error term reduces to:

$$|qu - p| = |x\delta_n| = x|\delta_n|.$$

If $x = 1$, then $q = q_n$, $p = p_n$, and we recover the equality $|qu - p| = |\delta_n|$. If $x \geq 2$, then $|qu - p| \geq 2|\delta_n| > |\delta_n|$, fulfilling the strict inequality.

Combining both cases concludes that $|qu - p| \geq |\delta_n|$ strictly holds for all $q \in (0, q_{n+1})$ unless $(p, q) = (p_n, q_n)$. ■

Definition A.5. An irrational number $u = [a_1, a_2, \dots] \in (0, 1) \setminus \mathbb{Q}$ is said to be of *bounded type* if its partial quotients are bounded, i.e. $\sup_{n \geq 1} a_n < \infty$.

The next result explains the fact that irrational numbers of bounded type cannot be approximated very well by rationals.

Proposition A.6. *Let $u = [a_1, a_2, \dots] \in (0, 1) \setminus \mathbb{Q}$. If u is of bounded type, then there exists some $\varepsilon > 0$ such that*

$$\left| u - \frac{p}{q} \right| \geq \frac{\varepsilon}{q^2}$$

for all rational numbers $\frac{p}{q}$.

Proof. Let $M = \sup_{n \geq 1} a_n$. Then

$$q_{n+1} \leq (M+1)q_n, \quad n \geq 0.$$

For any $q \in \mathbb{N}$, there is some n with $q_{n-1} < q \leq q_n$. Therefore

$$\left| u - \frac{p}{q} \right| > \left| u - \frac{p_n}{q_n} \right| \geq \frac{1}{q_n q_{n+2}} > \frac{1}{(M+1)^4 q^2}$$

as we wanted. ■

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